

Опр (кусочно-непрерывной ф-ции)

Ф-ция $f: [a, b] \rightarrow \mathbb{R}$ наз. КНФ, если она непрер. на $[a, b]$ за исключением конечного числа точек, которые явл-ся точками разрыва I^{∞} рода.

① КНФ на $[a, b]$ интер-на на $[a, b]$

Δ $f(x)$ - КНФ на $[a, b] \xRightarrow{\text{def}}$

$$\begin{cases} f(x) \in C^0([a, b] \setminus \{c_1, c_2, \dots, c_m\}), & c_i \in [a, b] \\ \exists f(c_i \pm 0) < +\infty, & i = \overline{1, m} \end{cases}$$

$$\begin{array}{c} | \quad | \quad | \quad | \quad | \\ a \quad c_1 \quad c_2 \quad \dots \quad c_m \quad b \end{array} \quad f(x) \in C^0([a, c_1], [c_1, c_2], \dots, [c_i, c_{i+1}], \dots, [c_m, b])$$

f явл-ся ст непрерывной ф-цией на

$\forall$ из от-в не более, чем в точке c_i

$$\Rightarrow f \in R([a, c_1], [c_i, c_{i+1}], [c_m, b], i = \overline{1, m})$$

$$\Rightarrow f \in R[a, b] \quad \Delta$$

Зам. зам. ① верно и в том случае, если $x=a, x=b$ - т.р I рода.

① Арифметические св-ва интер-х ф-ций
 $\int f, g \in R[a, b] \Rightarrow$

$$\textcircled{1} \quad \alpha f + \beta g \in R[a, b], \quad \alpha, \beta \in \mathbb{R}$$

$$\textcircled{2} \quad f \cdot g \in R[a, b]$$

$$\textcircled{3} \quad |f| \in R[a, b]$$

$$\textcircled{4} \quad \text{even } |f| > C \text{ on } [a, b], \quad C > 0$$

$$\Rightarrow \frac{1}{f} \in R[a, b]$$

$$\Delta 1) \quad \exists f, g \in R[a, b] \Rightarrow \overset{\text{u. Kr. D.}}{\int \lim_{\lambda(\tau) \rightarrow 0} \sum_{i=1}^n \omega(f, \Delta_i) \Delta x_i = 0}$$

$$\left\{ \begin{array}{l} \lim_{\lambda(\tau) \rightarrow 0} \sum_{i=1}^n \omega(g, \Delta_i) \Delta x_i = 0 \end{array} \right.$$

$$\Rightarrow \forall \varepsilon > 0 \exists \delta_1 = \delta_1(\varepsilon) > 0, \quad \forall \tau : \left\{ \begin{array}{l} \lambda(\tau) < \delta_1 \\ \lambda(\tau) < \delta_2 \end{array} \right.$$

$$\left\{ \begin{array}{l} \delta_2 = \delta_2(\varepsilon) > 0 \end{array} \right.$$

$$\Rightarrow \left\{ \begin{array}{l} \sum_{i=1}^n \omega(f, \Delta_i) \Delta x_i < \varepsilon \\ \sum_{i=1}^n \omega(g, \Delta_i) \Delta x_i < \varepsilon \end{array} \right. \quad \delta = \min(\delta_1, \delta_2) \quad \forall \tau : \lambda(\tau) < \delta$$

$$\textcircled{9} \quad | \alpha f(x) + \beta g(x) - \alpha f(y) - \beta g(y) | \leq$$

$$\leq |\alpha| |f(x) - f(y)| + |\beta| |g(x) - g(y)| \leq$$

$$\leq |\alpha| \omega(f, E) + |\beta| \omega(g, E); \quad \xrightarrow{\text{sup}}$$

$$\omega(\alpha f + \beta g, E) \leq |\alpha| \omega(f, E) + |\beta| \omega(g, E)$$

$\forall E.$

$$\sum_{i=1}^n \omega(\alpha f + \beta g, \Delta_i) \Delta x_i \leq |\alpha| \sum_{i=1}^n \omega(f, \Delta_i) \Delta x_i +$$

$$+ |\beta| \sum_{i=1}^n \omega(g, \Delta_i) \Delta x_i < |\alpha| \varepsilon + |\beta| \varepsilon = \tilde{\varepsilon}$$

$$\Rightarrow \alpha f + \beta g \in R[a, b] \quad \textcircled{a}$$

$$\textcircled{2} \quad \exists f, g \in R[a, b] \xrightarrow{\text{H. 4}} f, g - \text{ср на } [a, b]$$

$$\Rightarrow \exists C > 0: \begin{cases} |f| \leq C \\ |g| \leq C \end{cases} \quad \forall x \in [a, b]$$

$$\circ \quad |f(x)g(x) - f(y)g(y)| =$$

$$= |f(x)g(x) - f(x)g(y) + f(x)g(y) - f(y)g(y)| \leq$$

$$\leq |f(x)| |g(x) - g(y)| + |g(y)| |f(x) - f(y)|$$

$$\leq C (\omega(g, E) + \omega(f, E)) \xrightarrow{\text{суп л. 2.}}$$

$$\Rightarrow \omega(fg, E) \leq C (\omega(g, E) + \omega(f, E))$$

$$\Rightarrow \sum_{i=1}^n \omega(fg, \Delta_i) \Delta x_i \leq C \left(\sum_{i=1}^n \omega(g, \Delta_i) \Delta x_i + \sum_{i=1}^n \omega(f, \Delta_i) \Delta x_i \right) \quad \forall E$$

$$\leq C (\varepsilon + \varepsilon) = \varepsilon \xrightarrow{\text{л. Кр. Д}} fg \in R[a, b] \quad \textcircled{b}$$

$$\textcircled{3} \quad \circ \quad ||f(x)| - |f(y)|| \leq |f(x) - f(y)| \leq$$

$$\leq \sup_{x, y \in E} |f(x) - f(y)| = \omega(f, E)$$

$$\xrightarrow{\text{суп л. 2.}} \omega(|f|, E) \leq \omega(f, E) \quad \forall E$$

$$\Rightarrow \sum_{i=1}^n \omega(|f|, \Delta_i) \Delta x_i \leq \sum_{i=1}^n \omega(f, \Delta_i) \Delta x_i < \varepsilon$$

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$$\Rightarrow |f| \in R[a, b] \quad \textcircled{c}$$

$$\begin{aligned}
 (4) \quad & \left| \frac{1}{f(x)} - \frac{1}{f(y)} \right| = \left| \frac{f(y) - f(x)}{f(x)f(y)} \right| \leq \frac{|f(y) - f(x)|}{c^2} \leq \\
 & \leq \frac{1}{c^2} \sup_{x, y \in E} |f(y) - f(x)| = \frac{\omega(f, E)}{c^2} \\
 & \Rightarrow \omega\left(\frac{1}{f}, E\right) \leq \frac{\omega(f, E)}{c^2} \\
 & \sum_{i=1}^n \omega\left(\frac{1}{f}, \Delta_i\right) \Delta x_i \leq \sum_{i=1}^n \frac{\omega(f, \Delta_i)}{c^2} \Delta x_i < \frac{\varepsilon}{c^2} = \tilde{\varepsilon} \\
 & \Rightarrow \frac{1}{f} \in R[a, b] \quad \triangle
 \end{aligned}$$

(T₁) (линейность интеграла Римана)

$$\exists f, g \in R[a, b] \Rightarrow$$

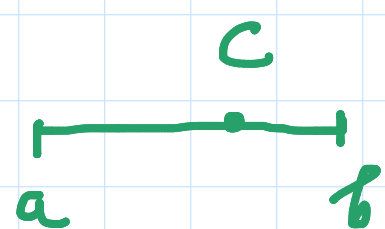
$$\int_a^b (\alpha f + \beta g) dx = \alpha \int_a^b f dx + \beta \int_a^b g dx$$

$$\triangle \text{ по } (T) \Rightarrow \alpha f + \beta g \in R[a, b]$$

$$\begin{aligned}
 \circ \int_a^b (\alpha f + \beta g) dx &= \lim_{\substack{\lambda(x) \rightarrow 0 \\ (n \rightarrow \infty)}} \sum_{i=1}^n (\alpha f(\xi_i) + \beta g(\xi_i)) \Delta x_i \\
 &= \alpha \lim_{\lambda(x) \rightarrow 0} \sum_{i=1}^n f(\xi_i) \Delta x_i + \beta \lim_{\lambda(x) \rightarrow 0} \sum_{i=1}^n g(\xi_i) \Delta x_i = \\
 &= \alpha \int_a^b f dx + \beta \int_a^b g dx \quad \triangle
 \end{aligned}$$

(T₂) (аддитивность интеграла Римана)

$$\exists f \in R[a, b], \quad c \in [a, b] \Rightarrow$$



$$\int_a^b f dx = \int_a^c f dx + \int_c^b f dx$$

$$\Delta \quad \exists f \in R[a, b] \Rightarrow \begin{cases} f \in R[a, c] \\ f \in R[c, b] \end{cases}$$

$\exists \tau$ — continuous $\tau: [a, b] \rightarrow \mathbb{R}$.

$$\exists \tau_1: [a, c], \quad \exists \tau_2: [c, b]$$

$$\begin{cases} \lambda(\tau_1) \leq \lambda(\tau) \\ \lambda(\tau_2) \leq \lambda(\tau) \end{cases}$$

$$\circ \sum_{i=1}^n f(\xi_i) \Delta x_i = \sum_{i=1}^n f(\xi_i) \Delta x_i + \sum_{i=1}^n f(\xi_i) \Delta x_i$$

$[a, b] \quad [a, c] \quad [c, b]$

$$\exists \lambda(\tau) \rightarrow 0 \Rightarrow \begin{cases} \lambda(\tau_1) \rightarrow 0 \\ \lambda(\tau_2) \rightarrow 0 \end{cases} \quad \text{by } \lim_{\lambda(\tau) \rightarrow 0} \lambda(\tau) = 0$$

$$\lim_{\lambda(\tau) \rightarrow 0} \sum_{i=1}^n f(\xi_i) \Delta x_i = \lim_{\lambda(\tau_1) \rightarrow 0} \sum_{i=1}^n f(\xi_i) \Delta x_i + \lim_{\lambda(\tau_2) \rightarrow 0} \sum_{i=1}^n f(\xi_i) \Delta x_i$$

$[a, b] \quad [a, c] \quad [c, b]$

$$\int_a^b f dx = \int_a^c f dx + \int_c^b f dx \quad \Delta$$

Example $\exists f \in R[\min(a, b, c), \max(a, b, c)]$

$$\Rightarrow \int_a^b f dx = \int_a^c f dx + \int_c^b f dx$$

$$\Delta \quad \int_a^b f dx = \int_a^c f dx + \int_c^b f dx \Rightarrow \int_a^b f dx = \int_a^c f dx - \int_b^c f dx = \int_a^c f dx + \int_c^b f dx \quad \Delta$$

Th

$$\exists f, g \in R[a, b], \quad a \leq b$$

$$\exists f(x) \leq g(x), \quad x \in [a, b] \Rightarrow$$

$$\int_a^b f dx \leq \int_a^b g dx$$

$$\Delta \quad \textcircled{1} \quad \sum_{i=1}^n f(\xi_i) \Delta x_i \leq \sum_{i=1}^n g(\xi_i) \Delta x_i \stackrel{\lim_{n \rightarrow \infty}}{=} \quad$$

$$\lim_{n \rightarrow \infty} \sum_{i=1}^n f(\xi_i) \Delta x_i \leq \lim_{n \rightarrow \infty} \sum_{i=1}^n g(\xi_i) \Delta x_i$$

$$\int_a^b f dx \leq \int_a^b g dx \quad \Delta$$

Corollary

$$\textcircled{2} \quad f \in R[a, b], \quad a \leq b$$

$$m = \inf_{x \in [a, b]} f(x), \quad M = \sup_{x \in [a, b]} f(x) \Rightarrow$$

$$m(b-a) \leq \int_a^b f dx \leq M(b-a)$$

$$\Delta \quad \text{T.v.} \quad m \leq f(x) \leq M, \quad x \in [a, b]$$

$$\int_a^b m dx \leq \int_a^b f(x) dx \leq \int_a^b M dx$$

$$m(b-a) \leq \int_a^b f dx \leq M(b-a) \quad \Delta$$

(T₄)

$$\textcircled{3} \quad f \in R[a, b] \Rightarrow$$

$$\left| \int_a^b f dx \right| \leq \int_a^b |f| dx$$